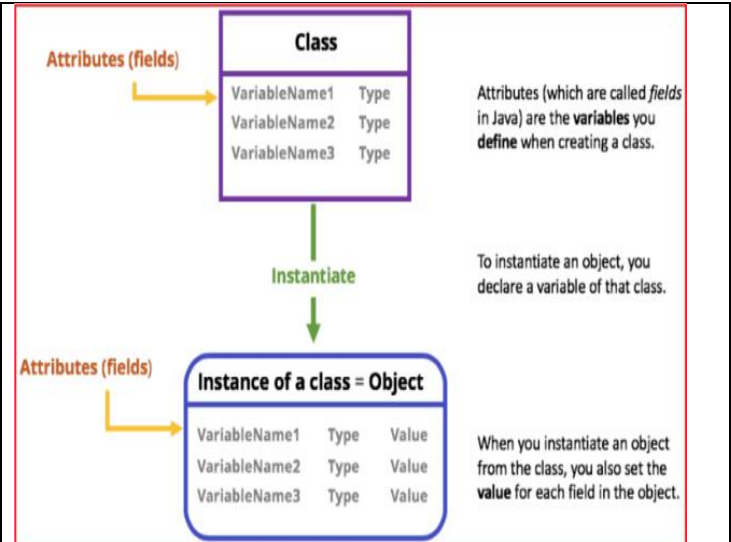


Class and Object

The Core Concept: Class and Object

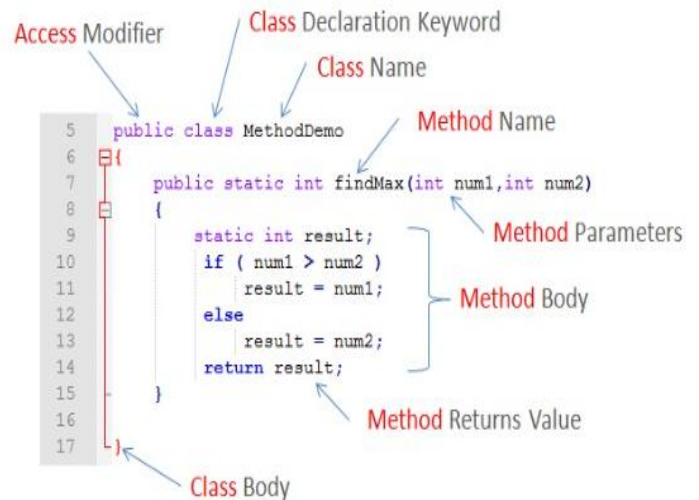
A Class is the Blueprint: It is the conceptual design, a template, or a factory mold. It does not physically exist; it just describes what something *will* look like and what it *will* do.

An Object is the Instance: It is the actual, physical item built directly from that blueprint. You can use one blueprint class to stamp out hundreds of individual objects

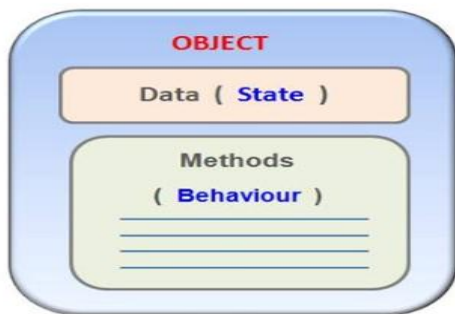


Anatomy of Class

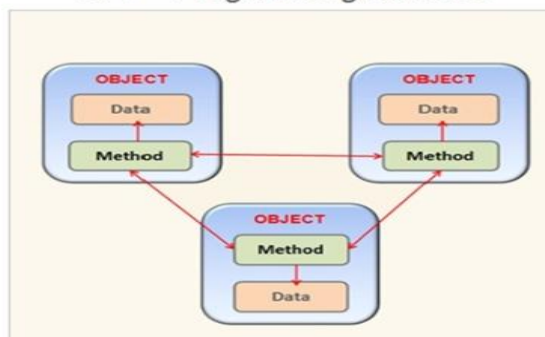
- **Attributes (Fields / Instance Variables):** What the object *knows* or remembers about itself (its data). Properties define the characteristics or attributes of an object.
- **Methods:** What the object can *do* (its functional actions).
- **Constructor:** The initialization engine that assigns starting values to an object's fields upon birth.



OOP - Concept Of Object



OPP - Program Organization



Example Blueprint

```
public class Car {  
    // Attributes  
    public String color;  
    public int speed;  
  
    // 1. THE CONSTRUCTOR (No return type, matches class name)  
    public Car(String initialColor, int initialSpeed) {  
        this.color = initialColor; // Sets the object's color to the incoming in  
        this.speed = initialSpeed; // Sets the object's speed to the incoming in  
    }  
  
    // Method  
    public void displayStatus() {  
        System.out.println("The " + color + " car is going " + speed + " mph.");  
    }  
}
```